

Baseline, Parametric, and Operator-Learning Surrogates for One-Dimensional Advection-Diffusion with Fixed Diffusion and Variable Advection

Ali Haghighi

Supervised by:

Afshin Ashrafzadeh, François Lehmann, Marwan Fahs

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Abstract

This report introduces a new one-dimensional advection–diffusion test family in which the physical time horizon and diffusion are fixed, while advection varies through the Courant–Friedrichs–Lewy (CFL) number. The formulation uses the normalized variables (x^*, t^*, C^*) and a fixed Péclet-type diffusive group, denoted here by Pe (here $Pe := DT_{\max}/L^2$, not the classical UL/D). The study considers $L = 100$ m, $T_{\max} = 1200$ d, $D = 10^{-6} \text{ m}^2 \text{ s}^{-1}$, and five velocities $U \in \{0.01, 0.02, 0.03, 0.04, 0.05\} \text{ m d}^{-1}$, corresponding to $CFL \in [0.12, 0.60]$ with $Pe = 0.010368$ fixed. The baseline PINN, parametric PINN, and parametric PINO are trained under a unified full-batch L-BFGS workflow on regular collocation meshes, differing only in architecture and in how CFL enters the model. Comparison with COMSOL reference solutions showed almost perfect agreement and confirmed that the Ogata–Banks solution remains appropriate in this regime, so it is retained as the analytical benchmark. The reported results show near-perfect agreement across all cases.

1 Introduction

This study has so far developed three physics-informed surrogates for one-dimensional contaminant transport: a baseline PINN, a parametric PINN, and a DeepONet-style PINO. In the earlier studies, improvements in architecture and training were often intertwined with changes in problem formulation, which made it harder to separate model behavior from setup-dependent effects. The present study shifts to a cleaner test family in which diffusion and the physical time horizon are fixed, and only the advective velocity varies.

Under this formulation, advection changes through the Courant–Friedrichs–Lewy (CFL) number while the diffusive group Pe remains fixed (again, here $Pe := DT_{\max}/L^2$, not UL/D). This provides a compact parameterization for the parametric models and places all three surrogates in the same normalized setting. It also leads to a regime in which the Ogata–Banks solution remains an appropriate analytical benchmark, consistent with the observed agreement with COMSOL reference solutions.

The baseline PINN, parametric PINN, and parametric PINO are therefore evaluated here under one common problem family, allowing the main differences to lie in the surrogate formulation itself: a single-regime network, a parameter-augmented PINN, and a branch–trunk operator model. This makes the present results a more direct view of how the three approaches behave when the governing setup is held fixed and advection is the only varying factor.

2 Problem Setup and Non-Dimensional Formulation

2.1 Physical advection–diffusion problem

We model one-dimensional solute transport in a saturated porous medium (or an equivalent one-dimensional continuum) with constant mean pore velocity $U > 0$ and constant hydrodynamic diffusion coefficient $D > 0$. The dependent variable $C(x, t)$ denotes solute concentration (mass per unit volume).

Advection appears through the term $U \partial C / \partial x$, and Fickian diffusion through $D \partial^2 C / \partial x^2$.

The governing *advection–diffusion equation* (ADE) is the linear second-order parabolic partial differential equation

$$\frac{\partial C}{\partial t} + U \frac{\partial C}{\partial x} = D \frac{\partial^2 C}{\partial x^2}, \quad (x, t) \in (0, L) \times (0, T_{\max}), \quad (1)$$

where $L > 0$ is a characteristic transport length (for example, column or domain length) and $T_{\max} > 0$ is the physical time horizon of interest (the maximum simulation time). Equation (1) states that the rate of change of concentration plus advective flux divergence equals the dispersive smoothing term. The symbols $\partial/\partial t$ and $\partial/\partial x$ denote partial derivatives with respect to physical time t and physical distance x , respectively.

2.2 Initial and boundary conditions

We impose a uniform zero initial condition,

$$C(x, 0) = 0, \quad x \in (0, L), \quad (2)$$

meaning the domain is initially clean. At the inlet $x = 0$ we prescribe a Dirichlet condition with fixed concentration $C_0 > 0$,

$$C(0, t) = C_0, \quad t \in (0, T_{\max}), \quad (3)$$

representing a constant upstream solute supply. At the outlet $x = L$ we impose

$$C(L, t) = 0, \quad t \in (0, T_{\max}), \quad (4)$$

corresponding to a clean downstream boundary. Together, (2)–(4) close the problem for (1). In applications where (4) is only an approximation, interpretation of solutions relative to semi-infinite references should be handled with care; here the conditions are stated as the mathematical model used in what follows.

2.3 Non-dimensional variables

We introduce dimensionless spatial and temporal coordinates scaled by L and $T_{\max}$, and a dimensionless concentration scaled by the inlet reference value C_0 :

$$x^* = \frac{x}{L}, \quad t^* = \frac{t}{T_{\max}}, \quad C^* = \frac{C}{C_0}. \quad (5)$$

Thus $x^* \in (0, 1)$ and $t^* \in (0, 1)$ when (x, t) ranges over the physical space–time rectangle. By the chain rule,

$$\frac{\partial}{\partial t} = \frac{1}{T_{\max}} \frac{\partial}{\partial t^*}, \quad \frac{\partial}{\partial x} = \frac{1}{L} \frac{\partial}{\partial x^*}, \quad \frac{\partial^2}{\partial x^2} = \frac{1}{L^2} \frac{\partial^2}{\partial x^{*2}}. \quad (6)$$

Substituting $C = C_0 C^*$ and using (6) in (1) gives

$$\frac{C_0}{T_{\max}} \frac{\partial C^*}{\partial t^*} + \frac{UC_0}{L} \frac{\partial C^*}{\partial x^*} = \frac{DC_0}{L^2} \frac{\partial^2 C^*}{\partial x^{*2}}. \quad (7)$$

Dividing (7) by C_0 yields the equivalent residual form with all terms moved to the left-hand side,

$$\frac{1}{T_{\max}} \frac{\partial C^*}{\partial t^*} + \frac{U}{L} \frac{\partial C^*}{\partial x^*} - \frac{D}{L^2} \frac{\partial^2 C^*}{\partial x^{*2}} = 0. \quad (8)$$

Multiplying (8) by $T_{\max}$ isolates a unit coefficient on the time derivative:

$$\frac{\partial C^*}{\partial t^*} + \frac{UT_{\max}}{L} \frac{\partial C^*}{\partial x^*} - \frac{DT_{\max}}{L^2} \frac{\partial^2 C^*}{\partial x^{*2}} = 0. \quad (9)$$

2.4 Scaled ADE and dimensionless groups

We define the two dimensionless groups

$$\text{CFL} := \frac{UT_{\max}}{L}, \quad \text{Pe} := \frac{DT_{\max}}{L^2}, \quad (10)$$

where CFL compares advective travel over the simulated horizon to the domain length, and Pe measures the strength of diffusion under the chosen $T_{\max}$ -based normalization (in contrast to the previous reports, where the Péclet number was defined in the classical form $\text{Pe} = UL/D$). The non-dimensional transport equation is therefore

$$\frac{\partial C^*}{\partial t^*} + \text{CFL} \frac{\partial C^*}{\partial x^*} - \text{Pe} \frac{\partial^2 C^*}{\partial x^{*2}} = 0. \quad (11)$$

with transformed conditions

$$C^*(x^*, 0) = 0, \quad C^*(0, t^*) = 1, \quad C^*(1, t^*) = 0, \quad (12)$$

3 Study Design

3.1 Fixed parameters, variable velocities, and resulting dimensionless coefficients

All three scripts share the same physical reference setup. The domain length is $L = 100$ m, the simulation horizon is $T_{\max} = 1200$ d, the reference concentration is $C_0 = 5$, and the diffusion coefficient is fixed at $D = 10^{-6} \text{ m}^2 \text{ s}^{-1} = 0.0864 \text{ m}^2 \text{ d}^{-1}$. The velocity is varied over

$$U \in \{0.01, 0.02, 0.03, 0.04, 0.05\} \text{ m d}^{-1},$$

which induces the corresponding CFL values $\{0.12, 0.24, 0.36, 0.48, 0.60\}$ while keeping $\text{Pe} = 0.010368$ fixed. Table 1 summarizes the regime.

Table 1: Physical setup and derived dimensionless coefficients used throughout the report.

Quantity	Value
Domain length L	100 m
Time horizon $T_{\max}$	1200 d
Inlet concentration C_0	5 kg m^{-3}
Diffusion D	$10^{-6} \text{ m}^2 \text{ s}^{-1} = 0.0864 \text{ m}^2 \text{ d}^{-1}$
Velocity set U	$\{0.01, 0.02, 0.03, 0.04, 0.05\} \text{ m d}^{-1}$
Derived CFL values	$\{0.12, 0.24, 0.36, 0.48, 0.60\}$
Fixed Pe	0.010368

3.2 Reference solution and evaluation strategy

Post-training evaluation is carried out through direct profile comparison with the Ogata–Banks analytical solution for semi-infinite one-dimensional transport under a constant inlet concentration [1]. In dimensional variables, this reference is

$$C(x, t) = \frac{C_0}{2} \left[\text{erfc} \left(\frac{x - Ut}{2\sqrt{Dt}} \right) + \exp \left(\frac{Ux}{D} \right) \text{erfc} \left(\frac{x + Ut}{2\sqrt{Dt}} \right) \right]. \quad (13)$$

It is used only for post-training comparison and never as a training target. Although the present model is posed on a finite domain with the outlet condition $C(L, t) = 0$, the parameter regime suggested that the Ogata–Banks reference should remain appropriate, since the plume remains upstream of the outlet at $x = L = 100$ m. To verify this, the Ogata–Banks profiles were compared with COMSOL numerical solutions for the same cases, and the agreement was essentially perfect; the corresponding figures are provided in Appendix A. The Ogata–Banks solution was therefore retained as the analytical benchmark for the reported comparisons.

4 Model Formulations

The baseline and parametric PINN use the standard physics-informed neural network framework, in which the governing PDE and constraints are enforced through automatic differentiation and a composite loss [2]. The third surrogate is a DeepONet-style operator model, which encodes the parameter and the space–time coordinates separately before combining their latent representations [3]. Table 2 summarizes the three models.

4.1 Baseline PINN

The baseline PINN solves a single fixed-CFL problem. It maps the input pair (x^*, t^*) to the dimensionless concentration $C^*(x^*, t^*) \in [0, 1]$. Its architecture is a fully connected MLP

$$[2, 16, 16, 16, 16, 16, 1],$$

with Tanh hidden activations, sigmoid output, and Xavier initialization.

4.2 Parametric PINN

The parametric PINN learns a family of solutions over varying advection strength by injecting CFL directly into the network input. The model takes (x^*, t^*, CFL) and returns $C^*(x^*, t^*, \text{CFL})$. Its architecture is the fully connected MLP

$$[3, 16, 16, 16, 16, 16, 1],$$

with Tanh hidden activations, sigmoid output, and Xavier-normal initialization. In this way the network learns a continuous parameterized solution manifold over the studied CFL interval.

Table 2: Model inputs, outputs, and architectural form.

Model	Inputs & output	Architecture
Baseline PINN	Input (x^*, t^*) ; output C^*	Fully connected MLP, width 16, Tanh hidden activations, sigmoid output
Parametric PINN	Input (x^*, t^*, CFL) ; output C^*	Fully connected MLP with architecture [3, 16, 16, 16, 16, 1]
Parametric PINO / DeepONet	Trunk input (x^*, t^*) , branch input CFL; output C^*	Branch [1, 16, 16, 16], trunk [2, 16, 16, 16], latent fusion plus sigmoid

4.3 Parametric PINO (DeepONet)

The PINO uses a DeepONet-style decomposition [3], with a branch network acting on the scalar parameter CFL and a trunk network acting on the coordinate pair (x^*, t^*) . The branch architecture is [1, 16, 16, 16], the trunk architecture is [2, 16, 16, 16], and their latent outputs are combined pointwise to produce the final prediction:

$$\hat{C}^*(x^*, t^*; \text{CFL}) := \sigma \left(\sum_{j=1}^p b_j(\text{CFL}) t_j(x^*, t^*) \right), \quad (14)$$

where b_j and t_j are the branch and trunk latent features, p is the latent width, and σ is the final sigmoid activation. Like the parametric PINN, this model learns C^* over varying CFL, but does so by separating parameter and coordinate encoding.

5 Training Workflow

All three models solve the same dimensionless ADE (11) on $(x^*, t^*) \in [0, 1]^2$ with the same initial and boundary conditions (12). The baseline model uses one fixed CFL value, whereas the parametric PINN and PINO treat CFL as an additional input dimension and learn a family of solutions over the full interval. The training protocol itself is intentionally aligned across the three scripts.

5.1 Collocation set construction

Training uses regular meshes with 50 points per axis and no random resampling. For the baseline PINN, the PDE interior collocation set is a 50×50 Cartesian grid in (x^*, t^*) , giving 2500 interior points. The initial condition uses 50 points along x^* at $t^* = 0$, and each boundary condition uses 50 points along t^* .

For the parametric PINN and PINO, the interior collocation set is a $50 \times 50 \times 50$ Cartesian grid in (x^*, t^*, CFL) , giving 125000 interior points. The initial-condition set is a 50×50 grid over (x^*, CFL) at $t^* = 0$, and the inlet and outlet sets are 50×50 grids over (t^*, CFL) . This produces fixed and fully repeatable coverage of the training domain. Figure 1 illustrates the baseline two-dimensional layout in (x^*, t^*) and the parametric three-dimensional layout in (x^*, t^*, CFL) ; for readability the panels plot a regular ten-point-per-axis mesh, whereas all training collocation sets use 50 points per axis as above.

5.2 Physics-informed objective

At every collocation point, automatic differentiation is used to compute the temporal derivative $\partial C^* / \partial t^*$, the spatial gradient

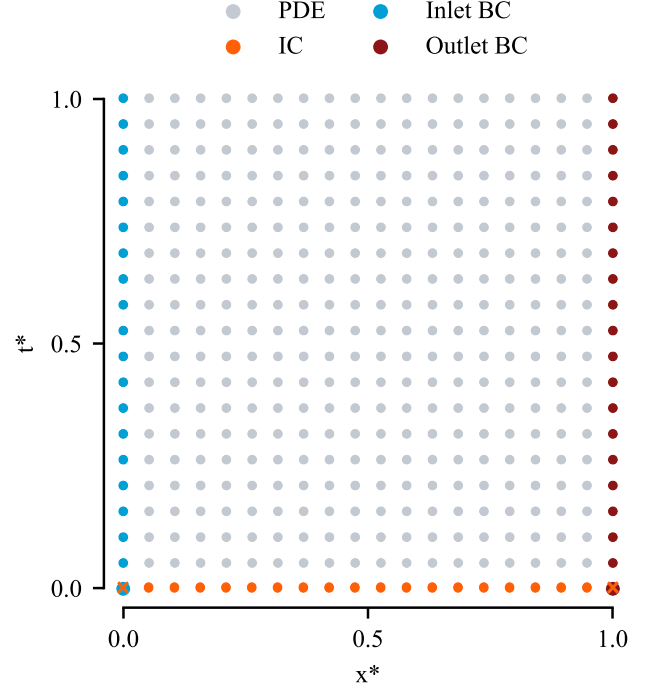
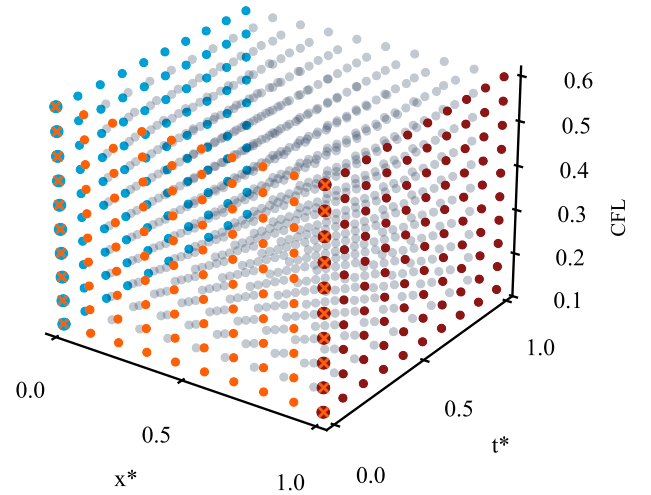
(a) Representation in (x^*, t^*) (b) Representation in (x^*, t^*, CFL)

Figure 1: Collocation layouts used in the study. The baseline PINN is trained on a regular two-dimensional (x^*, t^*) mesh, whereas the parametric PINN and PINO are trained on a regular three-dimensional (x^*, t^*, CFL) mesh. For readability these panels show a regular ten-point-per-axis plotted grid; all training runs use the 50-point-per-axis meshes described in the text.

$\partial C^*/\partial x^*$, and the spatial second derivative $\partial^2 C^*/\partial x^{*2}$. The residual is then assembled as

$$r = \frac{\partial C^*}{\partial t^*} + \text{CFL} \frac{\partial C^*}{\partial x^*} - \text{Pe} \frac{\partial^2 C^*}{\partial x^{*2}}, \quad (15)$$

with the pointwise CFL value used in the parametric models. The total loss is the equally weighted sum

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{inlet}} + \mathcal{L}_{\text{outlet}}, \quad (16)$$

where each term is a mean-squared error enforcing, respectively, the PDE residual, the zero initial condition, the inlet value $C^* = 1$, and the outlet value $C^* = 0$. This same loss structure is used in all three scripts.

5.3 Optimization with L-BFGS

All models are trained with full-batch L-BFGS for 1000 outer steps. The optimizer uses learning rate 1.0, history size 50, strong Wolfe line search, and `max_iter=1` per external step. Each closure evaluation computes the total physics-informed loss and its gradients on the fixed collocation mesh. The computations are performed in double precision (`float64`), and the same random seed is used across scripts for full reproducibility.

5.4 Unified view across the three models

All three models are trained under the same setup: the same non-dimensional PDE, the same deterministic collocation sets, the same four-term physics-informed loss, and the same full-batch L-BFGS optimization. The only differences are in the surrogate architecture and in how CFL enters the model: it is fixed in the baseline PINN, added directly as an input in the parametric PINN, and encoded through the branch network in the PINO.

6 Results

6.1 Baseline PINN

Figure 2 shows a baseline PINN trained for the fixed case $U = 0.05 \text{ m d}^{-1}$, corresponding to $\text{CFL} = 0.60$. The predicted concentration profiles are almost indistinguishable from the Ogata–Banks reference at all reported times. Fixed-CFL baseline runs for the other studied velocities are not shown here and are expected to give the same level of agreement.

6.2 Parametric PINN

Figure 3 shows the parametric PINN at three representative values spanning the studied range: $\text{CFL} = 0.12, 0.36$, and 0.60 . The model reproduces the concentration profiles with near-perfect agreement across low, intermediate, and high advection strength.

6.3 Parametric PINO (DeepONet)

Figure 4 shows the corresponding PINO results for the same three CFL values. The operator model likewise reproduces the solution family with near-perfect agreement.

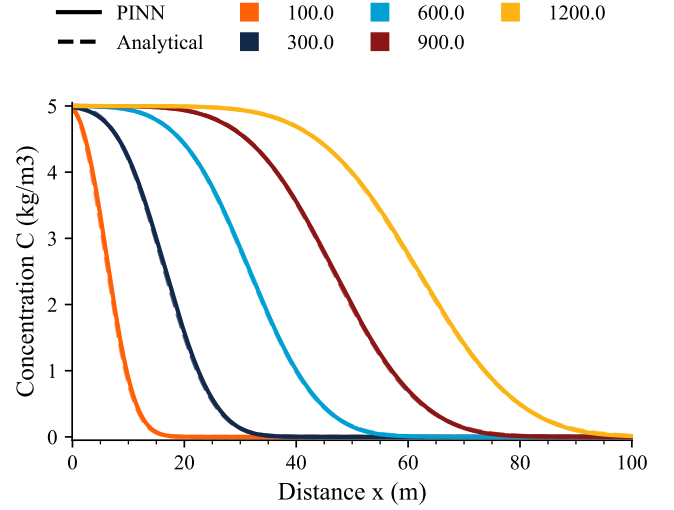


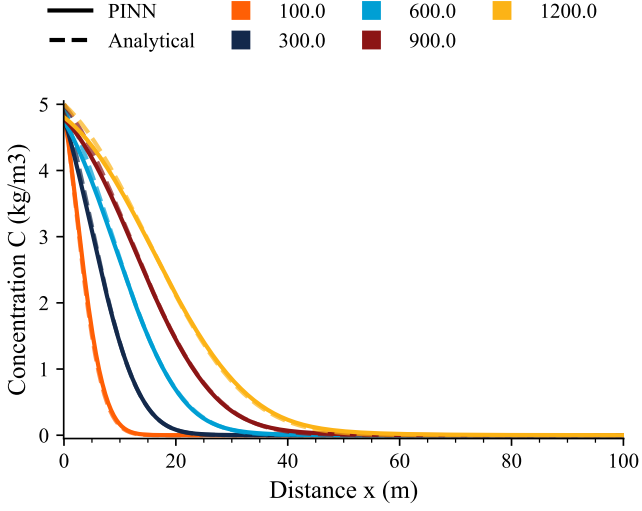
Figure 2: Baseline PINN concentration profiles for the fixed case $U = 0.05 \text{ m d}^{-1}$ ($\text{CFL} = 0.60$). The learned and reference profiles are nearly indistinguishable across the reported time slices.

7 Conclusion

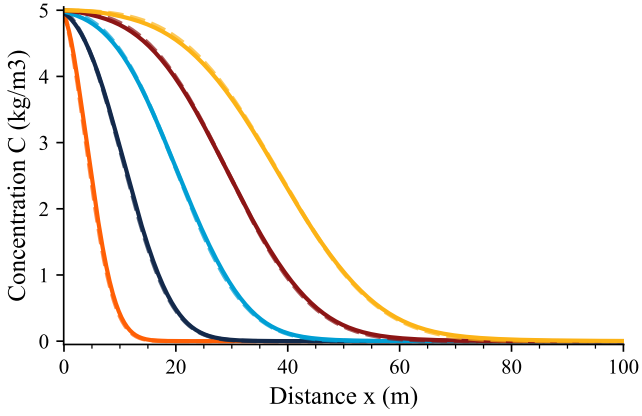
This report establishes a one-dimensional advection–diffusion test family in which diffusion and the physical time horizon are fixed, while advection varies through CFL. Under this formulation, the baseline PINN, parametric PINN, and DeepONet-style PINO were trained with the same deterministic full-batch L-BFGS workflow and all showed near-perfect agreement with the reference solution. The problem setup and training protocol are thus now aligned across the three surrogate classes. Since the present regime is already solved to near-perfect accuracy, a natural next direction is a more realistic transport setting, potentially with heterogeneous media and time- and space-varying input functions, where the neural-operator formulation can better exploit the kind of functional dependence it is designed to learn.

References

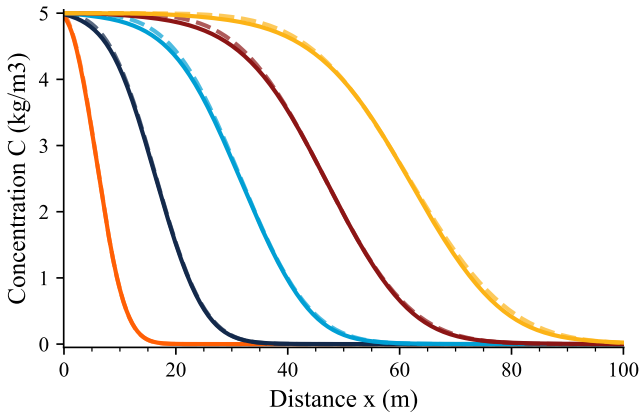
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(a) CFL = 0.12

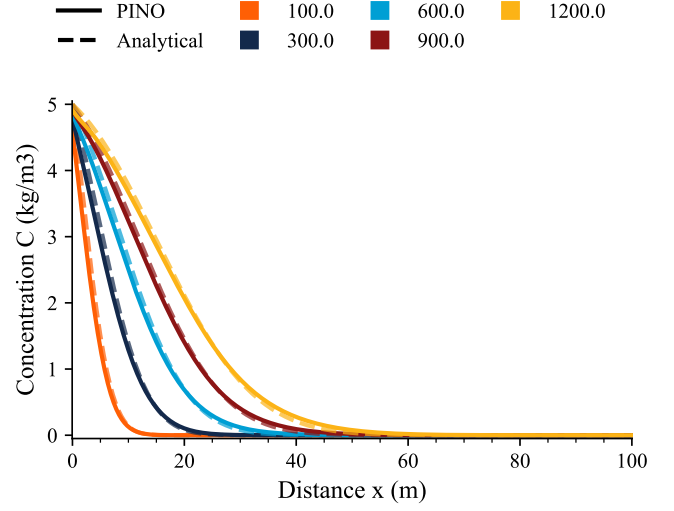


(b) CFL = 0.36

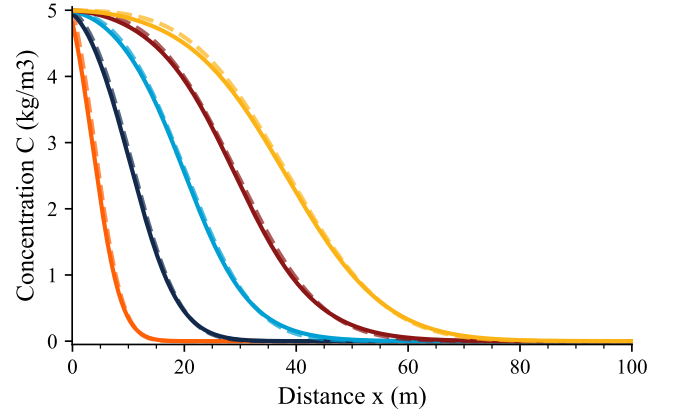


(c) CFL = 0.60

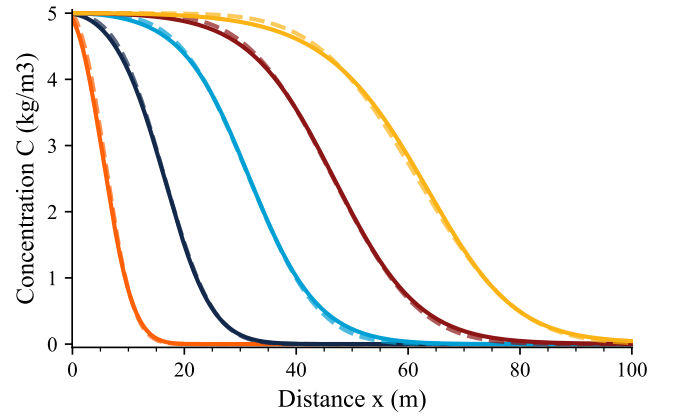
Figure 3: Parametric PINN concentration profiles at three representative values of CFL. In all cases, the predicted and analytical profiles are nearly indistinguishable.



(a) CFL = 0.12



(b) CFL = 0.36



(c) CFL = 0.60

Figure 4: Parametric PINO / DeepONet concentration profiles at three representative values of CFL. The predicted and analytical profiles are again nearly indistinguishable.

A Numerical Verification of the Ogata–Banks Reference

To verify that the Ogata–Banks solution is appropriate in the present regime, Figure A1 compares the analytical profiles with the numerical reference solutions provided for the five studied CFL values. The agreement is visually near-perfect in all cases, confirming that outlet effects remain negligible over the simulated time horizon and that Ogata–Banks is a suitable benchmark for the comparisons reported in this study.

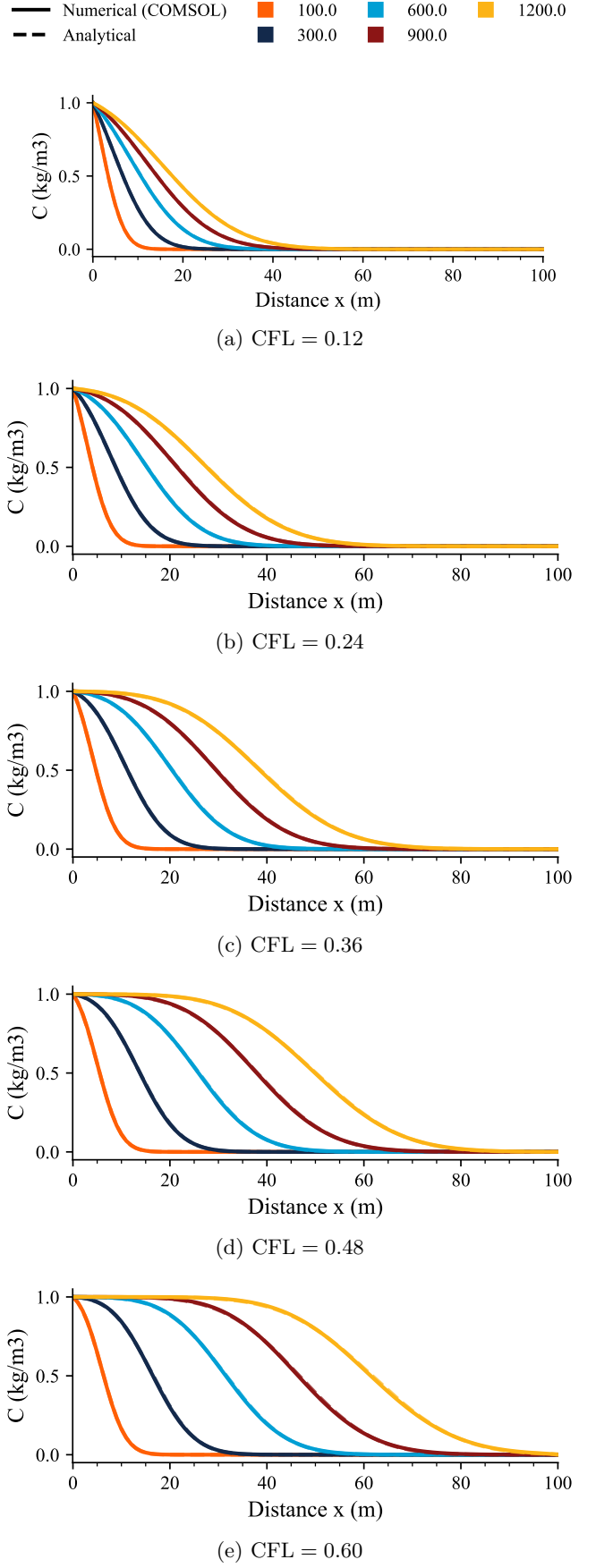


Figure A1: Comparison between the Ogata–Banks analytical solution and the numerical reference concentration profiles for the five studied CFL values.